

Aplicações Multimédia Interativas (AMI) Interactive Multimedia Applications (IMA) Spring 2025

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Who am I?

PhD in Computer Science, NOVA University of Lisbon, 2014

Postdoc at HIIT, University of Helsinki, 2013-2015

Assistant Professor, University of Madeira, 2016-2019

Research Fellow, ITI/LARSyS, IST, ULisboa, 2019-2025

Assistant Professor, ISEL, IPL, 2025-Present

Research in

Human-Computer Interaction (HCI), Multimedia,
Creativity Support Tools, Video Interactions, Pen and Touch
Interactions...

Course

2nd Semester

Hours/Week = 4hours: 1 theoretical-practical + 3 laboratory
(all in-person)

But most of the theoretical material will be at the beginning.

6 ECTS

Requirements: Programming Skills

Learning Outcomes

1. Know and understand the main user-centered design methods for developing mobile applications.
2. Know how to use hybrid platforms in the development of mobile applications.
3. Know how to apply user-centered methodologies in the development of mobile applications.
4. Develop critical thinking about the hybrid technologies used in the development of applications for different environments (e.g., Android or iOS).
5. Develop a mobile application that uses the main components usually included in a mobile application, e.g., sensor data

Evaluation

Grading

- 2 Lab Projects (LP): $0.5 * 1^{\text{st}} \text{ LP} + 0.5 * 2^{\text{nd}} \text{ LP}$

(LP min score 8)

Can be done in discussion with colleagues, but each student must submit his/her own version

- Final Project (Individual work): Report (R) + Application (A)

(R min score 8)

- Final grade = $0.35 * \text{LP} + 0.35 * \text{R} + 0.3 * \text{A}$

(min score 9.5) No exam but final discussion

Schedule

Tentative Dates (TBC)

1°Lab Project: Week 9-13 March (Ideation + Paper Prototype)

2°Lab Project: Week 13-17 April (Figma Prototype)

Final Project: Week 25-29 May – Discussions on the last week or after

Most of the work can be done during labs.

Technologies

We will study different technologies, as different ways to implement interactive multimedia applications:

- Flutter (Dart OOPL - C-style syntax - by Google)
- Unity (C#)
- QT (C++)

(all multi-platform)

You are free to choose the technology for your final project.

Bibliography

- Kasper Hornbæk, Per-Ola Kristensson, and Antti Oulasvirta (2025). *Introduction to Human-Computer Interaction*. Oxford University Press.
Open-Access: <https://introductiontohci.org/>
- Julie A. Jacko. 2012. *Human-Computer Interaction Handbook: Fundamentals, Evolving Technologies, and Emerging Applications (PART II – Computers in HCI)*, Third Edition (3rd ed.). CRC Press, Inc., Boca Raton, FL, USA.
- Jenny Preece, Helen Sharp and Yvonne Rogers, (2019), *Interaction Design: Beyond Human-Computer Interaction*, 5th Edition, John Wiley & Sons, Inc.
- Manuel J. Fonseca, Pedro Campos, Daniel Gonçalves, (2017), *Introdução ao Design de Interfaces*, 3a Edição, FCA.

Bibliography

- James D. Foley, Andries van Dam, Steven K. Feiner, and John F. Hughes. 1990. Computer Graphics: Principles and Practice (2nd Ed.). Addison-Wesley Longman Publishing Co., Inc., Boston, MA, USA.
- Alan Dix, Janet Finlay, Gregory Abowd, Russel Beale, (2004), "Human-Computer Interaction", 3rd Edition, Prentice Hall.
- Andrzej Marczewski, (2023), The Gamification Design Handbook: Even Ninja Monkeys Like to Play, Independently published.
- Mahesh Panhale, (2016), "Beginning Hybrid Mobile Application Development", Apress
- Robin Nixon, (2014), "Learning PHP, MySQL, JavaScript, CSS & HTML5", 3rd Edition, O'Reilly.

Tomorrow's Lecture: Introduction

Questions?